**Answer all questions**

**1** **(a)** State two differences between *random* and *systematic* errors.

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…… [2]

**(b)** Fig. 1.1 shows a simple pendulum consisting of a mass *m* attached to a light inextensible string of length *L*. The pendulum is secured to a fixed point and made to undergo oscillations when displaced sideways by a small angle *θ*.

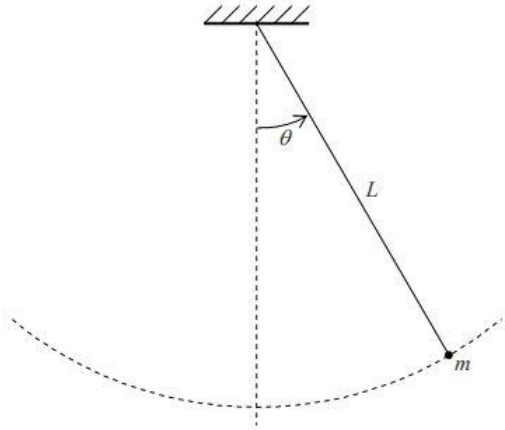


**Fig. 1.1 (not to scale)**

The period of oscillation *T* can be found by the following equation:

$$T^2 = 4\pi^2 \frac{L}{g}$$

where *g* is the acceleration of free fall.

**(i)** Given $L$ = 50.0 ± 0.2 cm; $g$ = 9.8 ± 0.1 m s$^{-2}$, find *T* with its associated uncertainty.

*T* = …… ± …… s [4]

For Examiner's Use

**(ii)** When determining the period of oscillation using a stopwatch, it is a good practice to take a large number of oscillations such that the total time taken is more than 20 seconds.

Explain why this should be done.

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..... [1]

**(iii)** The acceleration of free fall *g* on a body is caused by the gravitational field of the Earth.

Suggest why there is a variation in the strength of the Earth's gravitational field at different locations on Earth.

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..... [1]